JavaScript is disabled on your browser.

Java SE 27 & JDK 27

Class StringCharacterIterator

java.lang.Object
    java.text.StringCharacterIterator

All Implemented Interfaces:

`Cloneable`, `CharacterIterator`

```
public final class StringCharacterIterator
extends Object
implements CharacterIterator
```

`StringCharacterIterator` implements the `CharacterIterator` protocol for a `String`. The `StringCharacterIterator` class iterates over the entire `String`. All constructors throw `NullPointerException` if `text` is `null`.

Since:

1.1

See Also:

`CharacterIterator`

Field Summary

Fields declared in interface CharacterIterator
`DONE`

Constructor Summary

Constructors

Constructor	Description
`StringCharacterIterator (String text)`	Constructs an iterator with an initial index of 0.
`StringCharacterIterator (String text, int pos)`	Constructs an iterator with the specified initial index.
`StringCharacterIterator (String text, int begin, int end, int pos)`	Constructs an iterator over the given range of the given string, with the index set at the specified position.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
Object	clone()	Creates a copy of this iterator.
char	current()	Implements CharacterIterator.current() for String.
boolean	equals(Object obj)	Compares the equality of two StringCharacterIterator objects.
char	first()	Implements CharacterIterator.first() for String.
int	getBeginIndex()	Implements CharacterIterator.getBeginIndex() for String.
int	getEndIndex()	Implements CharacterIterator.getEndIndex() for String.
int	getIndex()	Implements CharacterIterator.getIndex() for String.
int	hashCode()	Computes a hashcode for this iterator.
char	last()	Implements CharacterIterator.last() for String.
char	next()	Implements CharacterIterator.next() for String.
char	previous()	Implements CharacterIterator.previous() for String.
char	setIndex(int p)	

		Implements CharacterIterator.setIndex() for String.
void	**setText**(String text)	Reset this iterator to point to a new string.

Methods declared in class Object

finalize, getClass, notify, notifyAll, toString, wait, wait, wait

Constructor Details

StringCharacterIterator

```
public StringCharacterIterator(String text)
```

Constructs an iterator with an initial index of 0.

Parameters:

`text` - the `String` to be iterated over

StringCharacterIterator

```
public StringCharacterIterator(String text,
                               int pos)
```

Constructs an iterator with the specified initial index.

Parameters:

`text` - The String to be iterated over

`pos` - Initial iterator position

Throws:

`IllegalArgumentException` - if `pos` is not within the bounds of range (inclusive) from `0` to the length of `text`

StringCharacterIterator

```
public StringCharacterIterator(String text,
                               int begin,
                               int end,
                               int pos)
```

Constructs an iterator over the given range of the given string, with the index set at the specified position.

Parameters:

`text` - The String to be iterated over

`begin` - Index of the first character

`end` - Index of the character following the last character

`pos` - Initial iterator position

Throws:

`IllegalArgumentException` - if `begin` and `end` are not within the bounds of range (inclusive) from `0` to the length of `text`, `begin` is greater than `end`, or `pos` is not within the bounds of range (inclusive) from `begin` to `end`

Method Details

setText

```
public void setText(String text)
```

Reset this iterator to point to a new string. This package-visible method is used by other java.text classes that want to avoid allocating new StringCharacterIterator objects every time their setText method is called.

Parameters:

`text` - The String to be iterated over

Throws:

`NullPointerException` - if `text` is `null`

Since:

1.2

first

```
public char first()
```

Implements CharacterIterator.first() for String.

Specified by:

`first` in interface `CharacterIterator`

Returns:

the first character in the text, or DONE if the text is empty

See Also:

`CharacterIterator.first()`

last

```
public char last()
```

Implements CharacterIterator.last() for String.

Specified by:

`last` in interface `CharacterIterator`

Returns:

the last character in the text, or DONE if the text is empty

See Also:

`CharacterIterator.last()`

setIndex

```
public char setIndex(int p)
```

Implements CharacterIterator.setIndex() for String.

Specified by:

`setIndex` in interface `CharacterIterator`

Parameters:

p - the position within the text. Valid values range from getBeginIndex() to getEndIndex(). An IllegalArgumentException is thrown if an invalid value is supplied.

Returns:

the character at the specified position or DONE if the specified position is equal to getEndIndex()

Throws:

`IllegalArgumentException` - if p is not within the bounds (inclusive) of `getBeginIndex()` to `getEndIndex()`

See Also:

`CharacterIterator.setIndex(int)`

current

```
public char current()
```

Implements CharacterIterator.current() for String.

Specified by:

`current` in interface `CharacterIterator`

Returns:

the character at the current position or DONE if the current position is off the end of the text.

See Also:

`CharacterIterator.current()`

next

```
public char next()
```

Implements CharacterIterator.next() for String.

Specified by:

`next` in interface `CharacterIterator`

Returns:

the character at the new position or DONE if the new position is off the end of the text range.

See Also:

`CharacterIterator.next()`

previous

```
public char previous()
```

Implements CharacterIterator.previous() for String.

Specified by:

`previous` in interface `CharacterIterator`

Returns:

the character at the new position or DONE if the current position is equal to getBeginIndex().

See Also:

`CharacterIterator.previous()`

getBeginIndex

```
public int getBeginIndex()
```

Implements CharacterIterator.getBeginIndex() for String.

Specified by:

`getBeginIndex` in interface `CharacterIterator`

Returns:

the index at which the text begins.

See Also:

`CharacterIterator.getBeginIndex()`

getEndIndex

```
public int getEndIndex()
```

Implements CharacterIterator.getEndIndex() for String.

Specified by:

`getEndIndex` in interface `CharacterIterator`

Returns:

the index after the last character in the text

See Also:

`CharacterIterator.getEndIndex()`

getIndex

```
public int getIndex()
```

Implements CharacterIterator.getIndex() for String.

Specified by:

`getIndex` in interface `CharacterIterator`

Returns:

the current index.

See Also:

`CharacterIterator.getIndex()`

equals

```
public boolean equals(Object obj)
```

Compares the equality of two StringCharacterIterator objects.

Overrides:

`equals` in class `Object`

Parameters:

`obj` - the StringCharacterIterator object to be compared with.

Returns:

true if the given obj is the same as this StringCharacterIterator object; false otherwise.

See Also:

`Object.hashCode()`, `HashMap`

hashCode

```
public int hashCode()
```

Computes a hashcode for this iterator.

Overrides:

hashCode in class Object

Returns:

A hash code

See Also:

Object.equals(java.lang.Object), System.identityHashCode(Object)

clone

```
public Object clone()
```

Creates a copy of this iterator.

Specified by:

clone in interface CharacterIterator

Overrides:

clone in class Object

Returns:

A copy of this

See Also:

Cloneable

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
Java is a trademark or registered trademark of Oracle and/or its affiliates in the US and other countries.
Copyright © 1993, 2026, Oracle and/or its affiliates, 500 Oracle Parkway, Redwood Shores, CA 94065 USA.
All rights reserved. Use is subject to license terms and the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.